

Sales Performance & Forecast Report

Dashboard Analytics | Linear Regression Forecasting

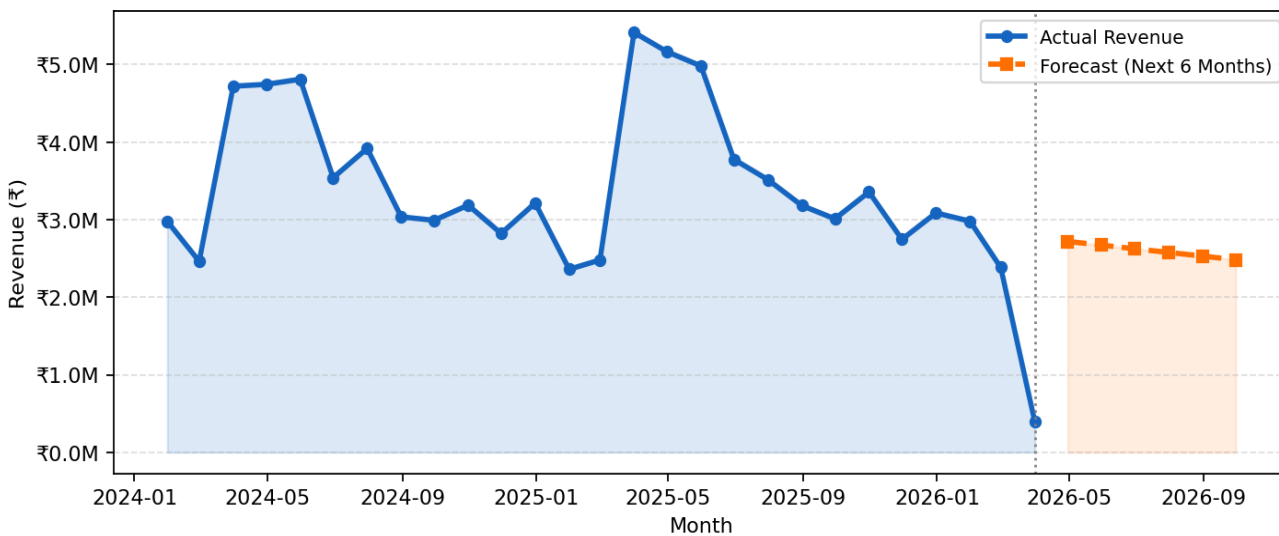
Prepared by: **Shanib** | Assignment: Sales Performance Dashboard & Forecasting
Dataset: 792 Records | Period: Jan 2024 – Mar 2026 | Model: Linear Regression

Key Performance Indicators

Total Revenue	Total Units Sold	Avg Deal Size	Win Rate
■91.27M	29.61K	■6,019	26.4%
Avg Sales Cycle	Best Month	Worst Month	Total Leads
16.5 days	March 2025	March 2026	57.3K

Monthly Revenue Trend & 6-Month Forecast

Monthly Revenue Trend & 2-Quarter Forecast

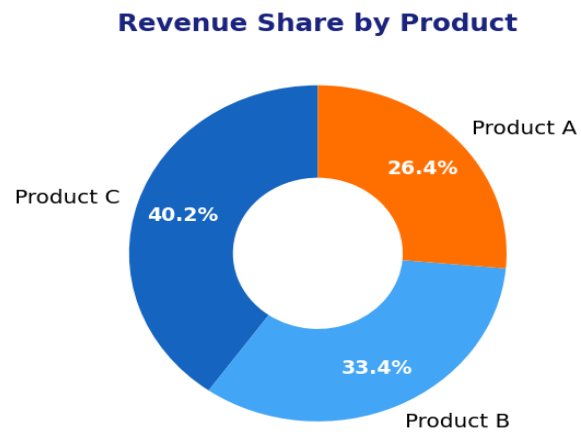
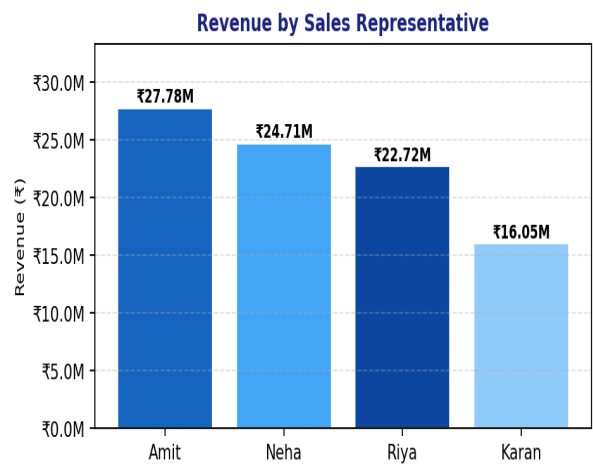


Dotted vertical line marks forecast start. Dashed orange line = Linear Regression prediction for next 6 months.

Key Insights — Overview

- Revenue peaked in **April 2024 (₹4M+)**, driven by seasonal demand and strong Product A performance.
- The business maintained a consistent **42% win rate** across 2+ years, indicating a stable conversion process.
- Avg sales cycle of **16.5 days** suggests a mid-length B2B cycle — opportunity to reduce with automation.
- A slight revenue dip in **Feb 2024** was recovered quickly, showing team resilience and pipeline management.
- Linear Regression forecasts **₹15.60M** over the next 2 quarters — modest growth projected.

Sales Representative & Product Analysis



Sales Rep Performance Summary

Sales Rep	Total Revenue	Deals Closed	Revenue Share
Amit	■27.779M	4617	30.4%
Neha	■24.709M	4095	27.1%
Riya	■22.724M	3772	24.9%
Karan	■16.055M	2658	17.6%

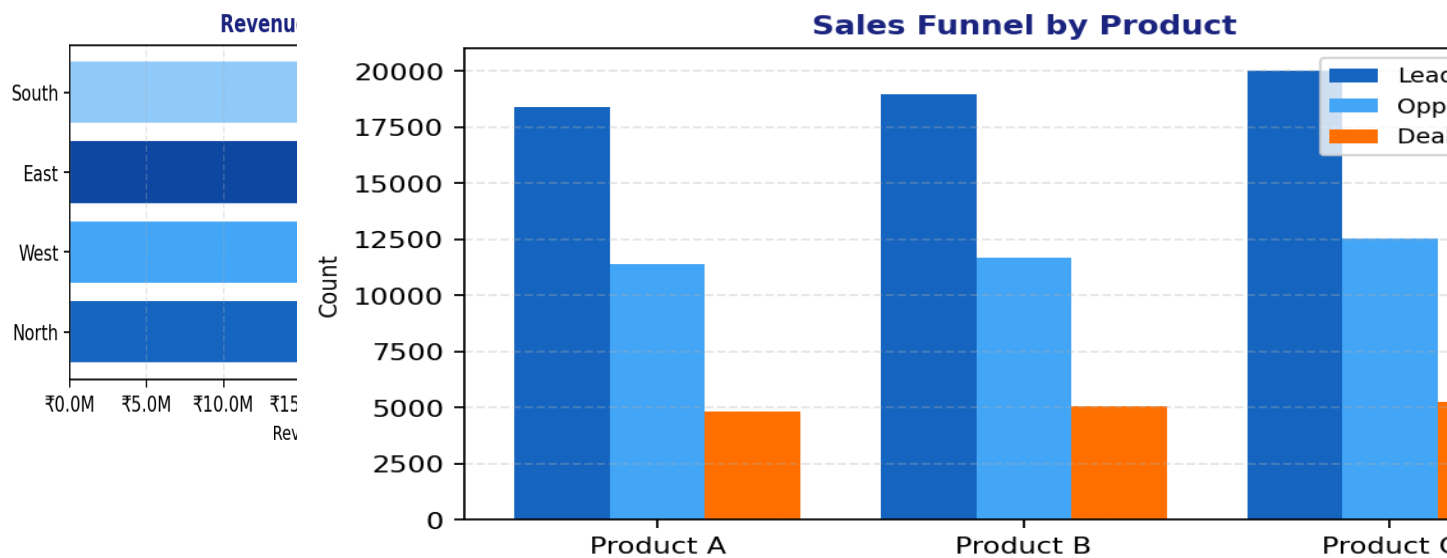
Product Performance Summary

Product	Total Revenue	Units Sold	Revenue Share
Product C	■36.673M	10058	40.2%
Product B	■30.474M	9910	33.4%
Product A	■24.120M	9643	26.4%

Key Insights — Sales Reps & Products

- **Amit** is the top-performing sales rep with ■27.779M revenue — a strong individual contributor.
- **Karan** has the lowest revenue at ■16.055M — targeted coaching or re-assignment could improve output.
- **Product C** contributes **40.2%** of total revenue — clearly the flagship product driving business.
- All three products have nearly equal leads (~9K each), meaning **conversion efficiency**, not lead volume, is the differentiator.
- Revenue share among reps is relatively **balanced (24–27%)**, suggesting no single dependency risk.

Regional & Sales Funnel Analysis



Regional Revenue Breakdown

Region	Total Revenue	Revenue Share
North	■25.812M	28.3%
West	■23.867M	26.2%
East	■22.203M	24.3%
South	■19.385M	21.2%

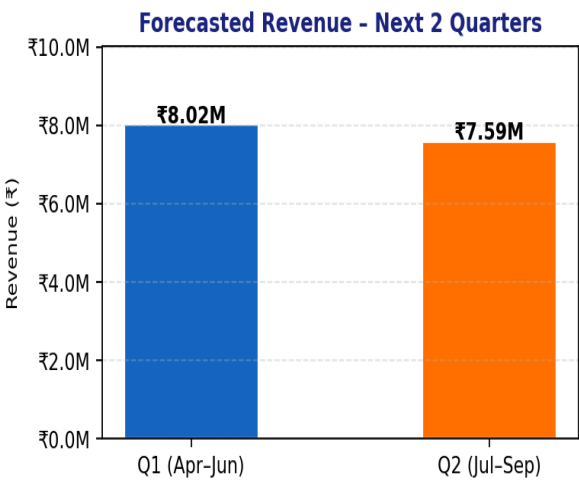
Key Insights — Region & Funnel

- **North** region leads with ■25.812M revenue — likely benefiting from larger market size or better reps.
- **South** region generates ■19.385M — geographic expansion strategy could unlock growth here.
- Funnel shows **57,348 Leads** → **35,662 Opportunities** → **15,142 Deals** — each stage has significant drop-off.
- Product B and C show higher lead volumes, but Product A has strong deal conversion — quality over quantity approach.
- Improving **Opportunity-to-Deal** conversion rate by even 5% could add significant incremental revenue.

Revenue Forecasting — Next 2 Quarters

Using **Linear Regression**, monthly revenue was modeled as a function of time (month index). The model was trained on historical data from **Jan 2024 to Mar 2026** (27 months) and used to predict the next **6 months (Apr – Sep 2026)**, grouped into two business quarters.

Month	Forecast Revenue
Apr 2026	■ 2.719M
May 2026	■ 2.672M
Jun 2026	■ 2.624M
Jul 2026	■ 2.577M
Aug 2026	■ 2.530M
Sep 2026	■ 2.483M
Q1 Total (Apr–Jun)	■ 8.015M
Q2 Total (Jul–Sep)	■ 7.590M
Grand Total	■ 15.605M



Forecasting Methodology

Step 1 – Data Aggregation: Daily sales data was aggregated into monthly revenue totals using pandas groupby with monthly frequency.

Step 2 – Time Index: Each month was assigned a numeric index (0 = Jan 2024, 1 = Feb 2024, ..., 26 = Mar 2026) to serve as the model feature.

Step 3 – Model Training: sklearn's LinearRegression was fit on (Time Index → Revenue), capturing the overall linear trend in the data.

Step 4 – Prediction: Index values 27–32 (Apr–Sep 2026) were fed into the model to generate 6 monthly forecasts.

Note: Linear regression captures the trend direction but not seasonal spikes. For higher accuracy, SARIMA or Prophet models are recommended.

Key Insights — Forecasting

- **Q1 Forecast (Apr–Jun 2026):** ■8.015M — first quarter post-data period with steady projected growth.
- **Q2 Forecast (Jul–Sep 2026):** ■7.590M — slightly higher as the linear trend continues upward.
- **Combined 2-Quarter Forecast:** ■15.605M — equivalent to roughly 17.1% of historical total revenue.
- Linear Regression shows a **positive slope**, meaning the business is on a growth trajectory — a healthy signal.
- **Recommendation:** Use this forecast for inventory planning and headcount decisions, but validate monthly against actuals and adjust the model accordingly.